

Immune-modulatory effects of dietary Yeast Beta-1,3/1,6-D-glucan

Beta-glucans are a heterogeneous group of natural polysaccharides mostly investigated for their immunological effects. Due to the low systemic availability of oral preparations, it has been thought that only parenterally applied beta-glucans can modulate the immune system. However, several in vivo and in vitro investigations have revealed that orally applied beta-glucans also exert such effects. Various receptor interactions, explaining possible mode of actions, have been detected. The effects mainly depend on the source and structure of the beta-glucans. In the meantime, several human clinical trials with dietary insoluble yeast beta-glucans have been performed. The results confirm the previous findings of in vivo studies. The results of all studies taken together clearly indicate that oral intake of insoluble yeast beta-glucans is safe and has an immune strengthening effect.